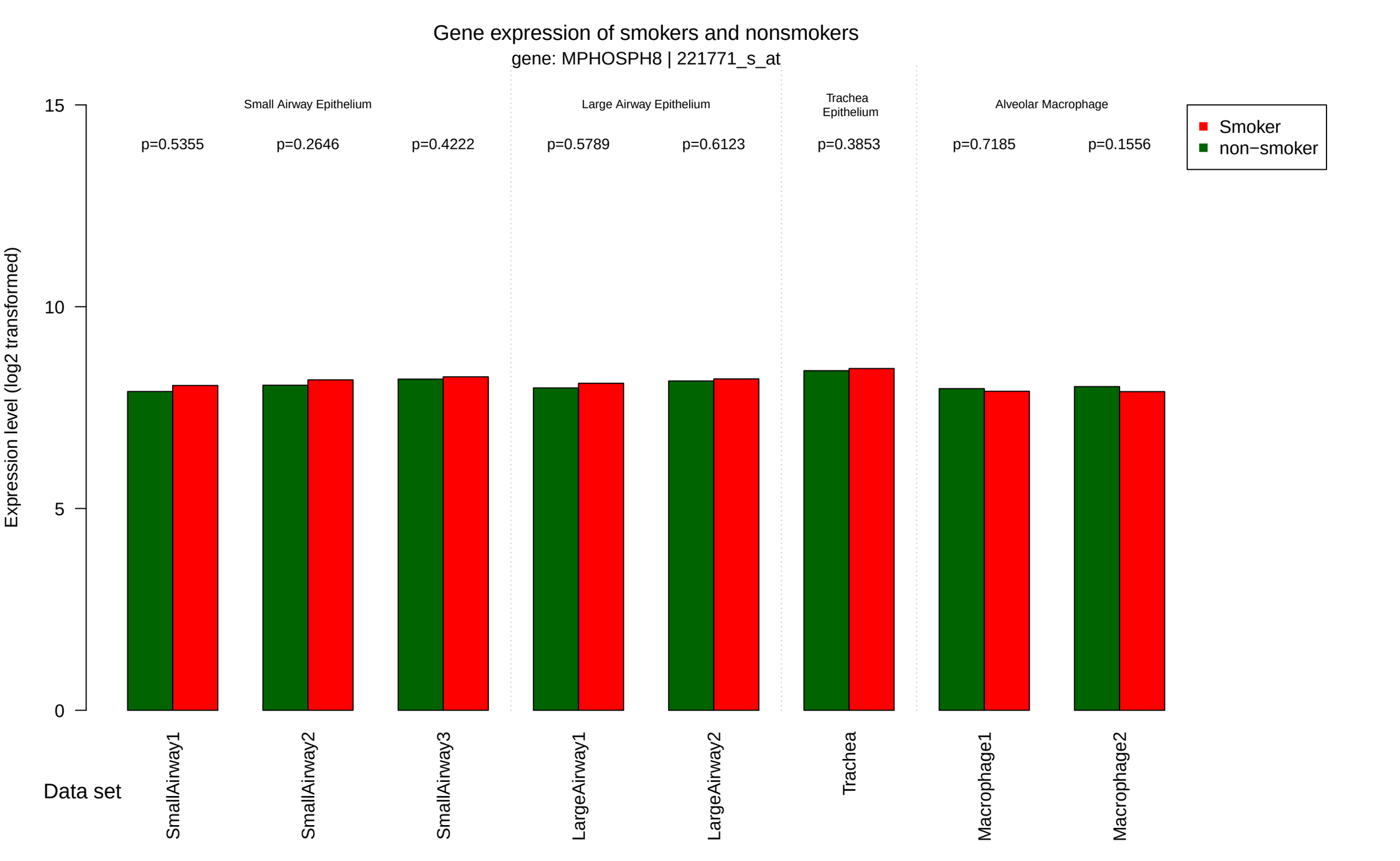
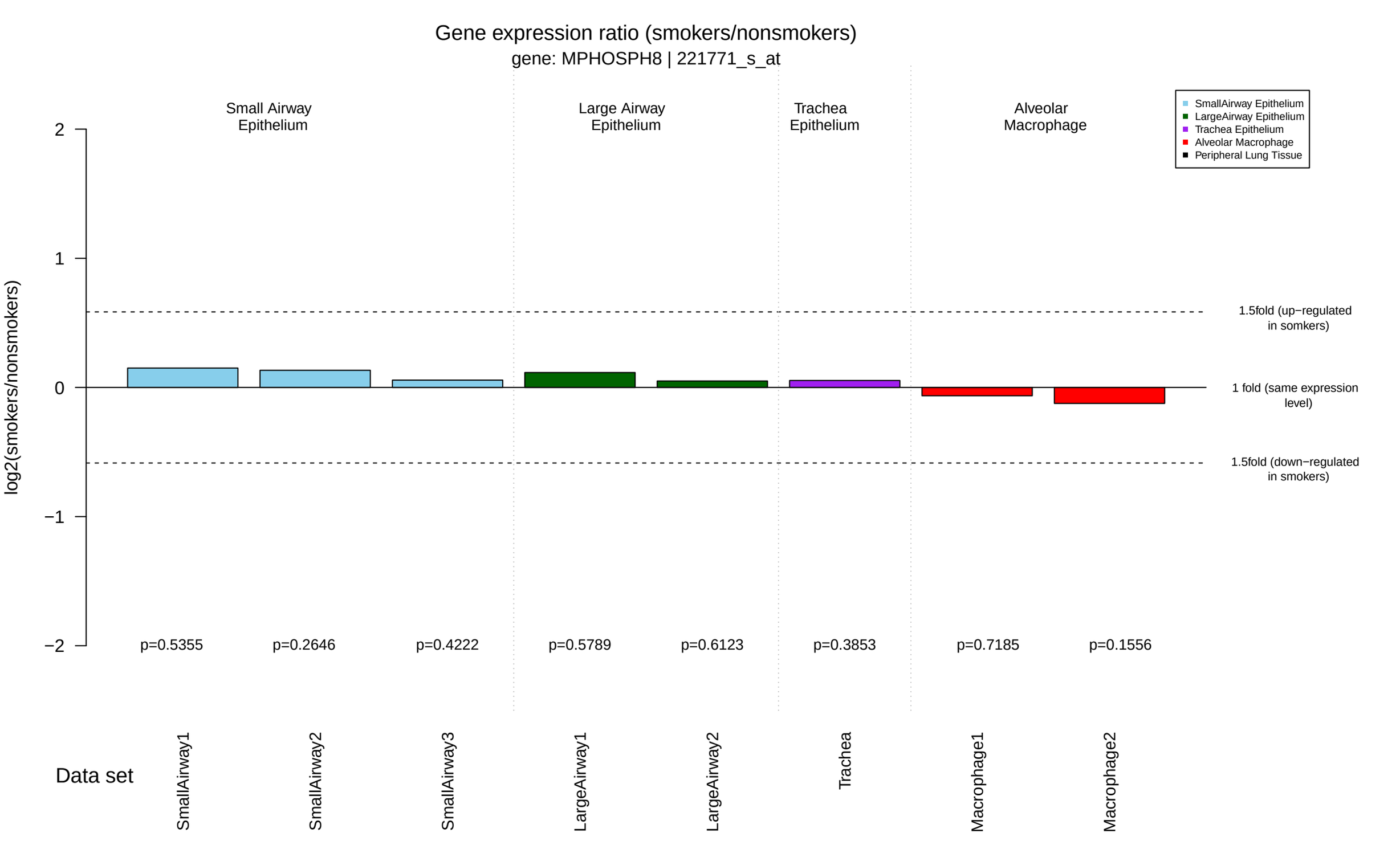
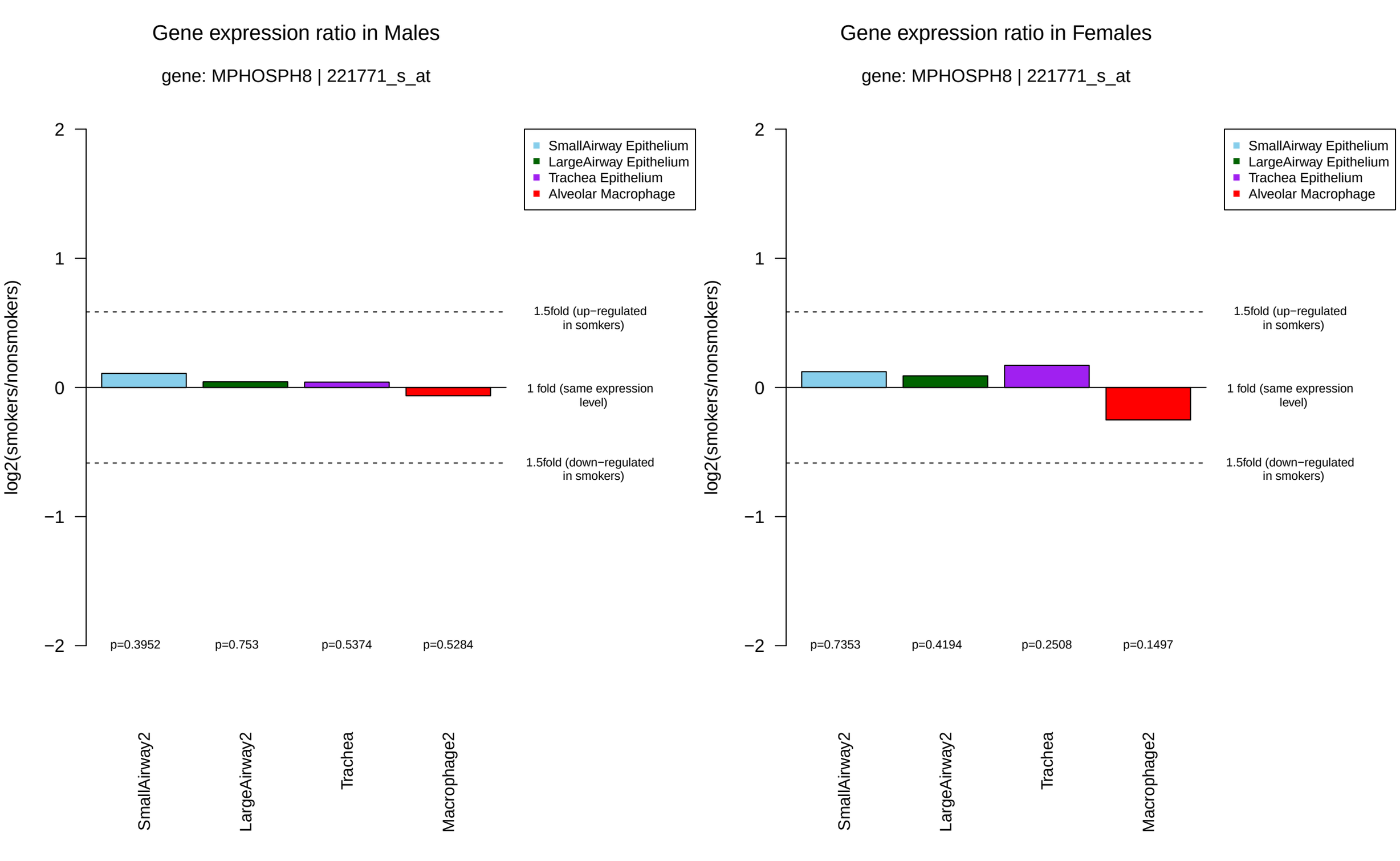
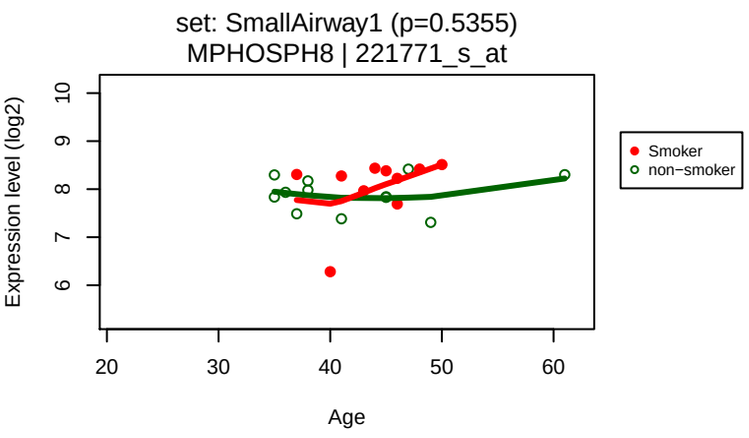
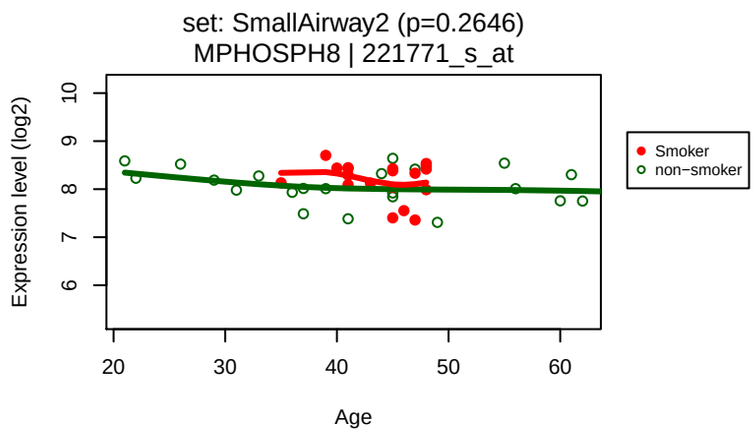
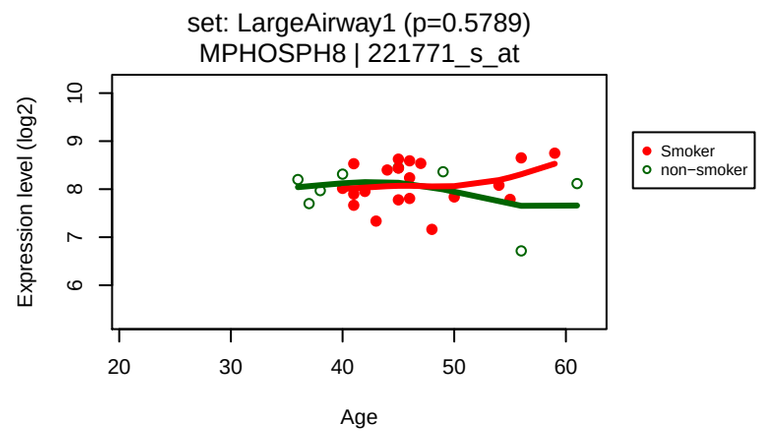

Gene expression of smokers and nonsmokers
gene: MPHOSPH8 | 221771_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5355
p=0.2646
p=0.4222
p=0.5789
p=0.6123
p=0.3853
p=0.7185
p=0.1556
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

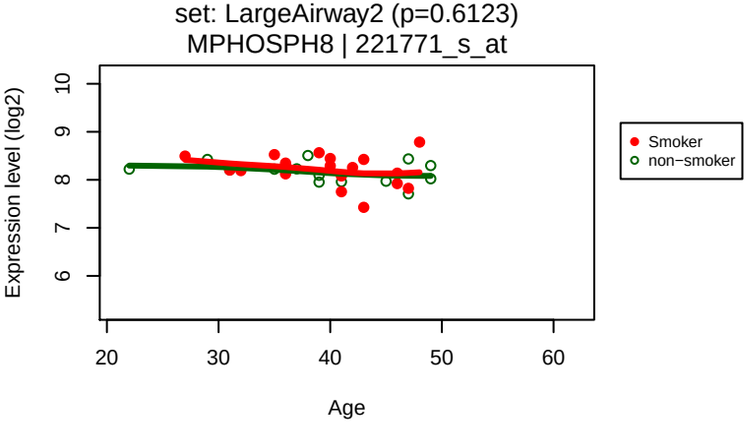

Gene expression ratio (smokers/nonsmokers)
gene: MPHOSPH8 | 221771_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5355
p=0.2646
p=0.4222
p=0.5789
p=0.6123
p=0.3853
p=0.7185
p=0.1556
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

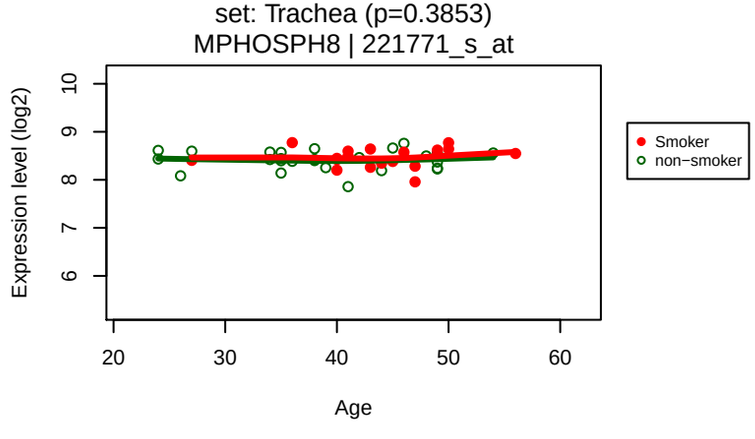

Gene expression ratio in Males
gene: MPHOSPH8 | 221771_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3952
p=0.753
p=0.5374
p=0.5284
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MPHOSPH8 | 221771_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7353
p=0.4194
p=0.2508
p=0.1497
SmallAirway2
LargeAirway2
Trachea
Macrophage2

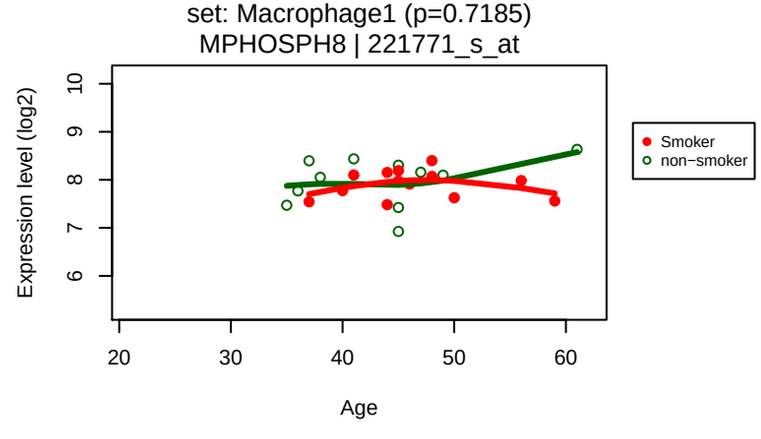

set: SmallAirway1 (p=0.5355)
MPHOSPH8 | 221771_s_at
Expression level (log2)
Age
Smoker
non−smoker

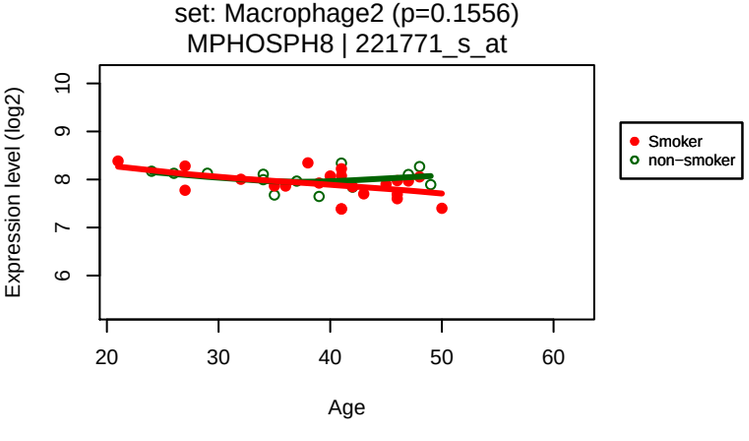

set: SmallAirway2 (p=0.2646)
MPHOSPH8 | 221771_s_at
Expression level (log2)
Age
Smoker
non−smoker

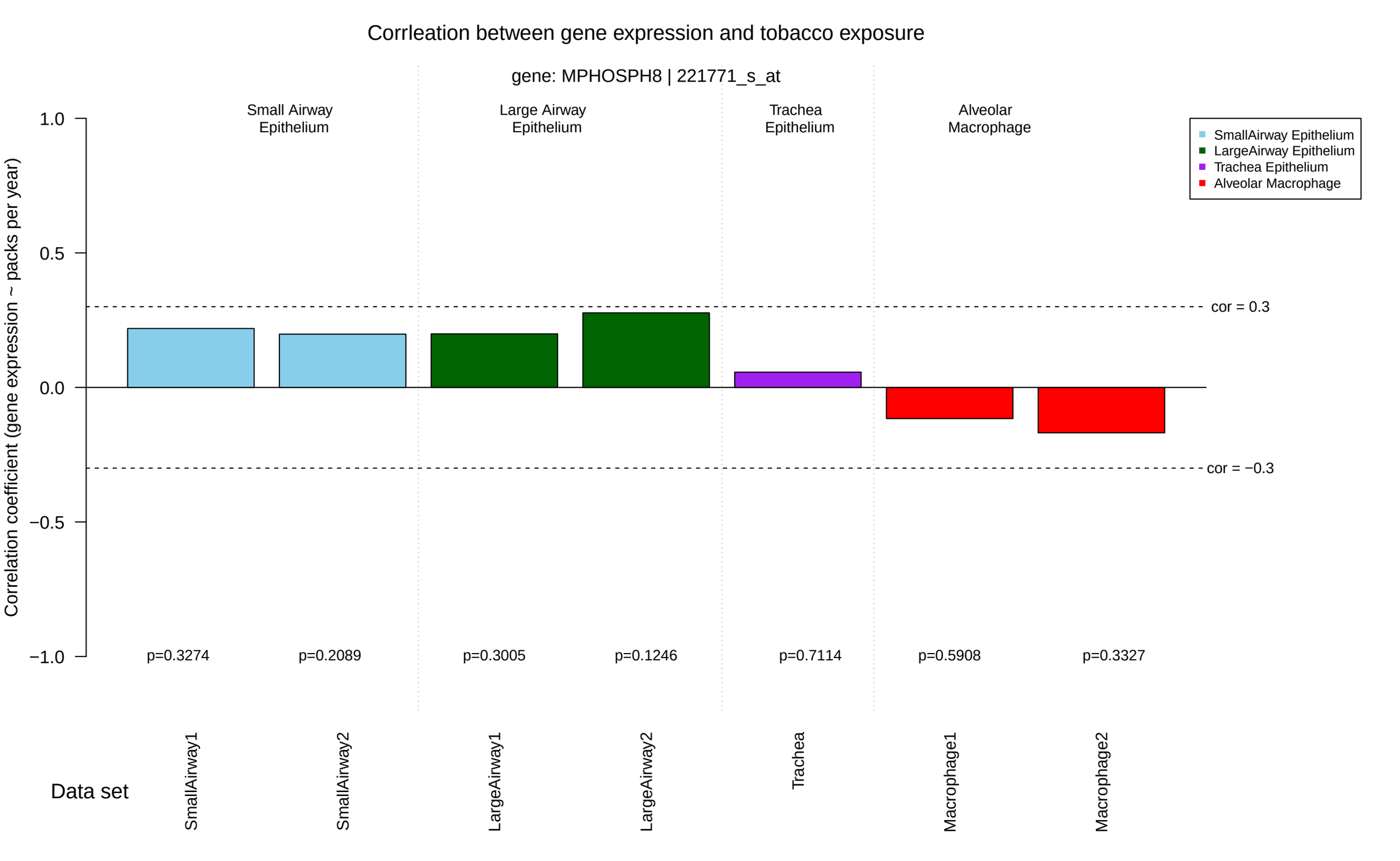

set: LargeAirway1 (p=0.5789)
MPHOSPH8 | 221771_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6123)
MPHOSPH8 | 221771_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3853)
MPHOSPH8 | 221771_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7185)
MPHOSPH8 | 221771_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1556)
MPHOSPH8 | 221771_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MPHOSPH8 | 221771_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3274
p=0.2089
p=0.3005
p=0.1246
p=0.7114
p=0.5908
p=0.3327
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2